MILITARY SPECIFICATION
SPECIAL HANDLING EQUIPMENT

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope. This specification covers the performance, design, and test requirements for the Special Handling Equipment (SHE).

1.2 Classification. The SHE shall consist of the following items (see 6.4).

a. Adapter, Handling, Antenna MX-9046( )/AWM-23(V)

b. Adapter, Handling, Tactical Information Display MX-9047( )/AWM-23(V)

c. Bar, Lifting, Antenna MX-9048( )/AWM-23(V)

d. Adapter, Hoisting, Antenna ADU-371( )/E

e. Adapter, Handling, Antenna MX-9691( )/AWM-23(V)

1.3 Associated equipment. The antenna handling adapter shall operate with Test Set, Antenna, Boresight TS-3568( )/AWM-23(V) (See 6.4).

2. APPLICABLE DOCUMENTS

2.1 General. The following documents, of the issue specified in AS-4053, Applicable Document List for AWCS Test Set AN/AWM-23(V), form a part of this specification to the extent specified herein. MIL-T-81734(AS) shall be the applicable general specification for the SHE described herein (see 6.6).

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SPECIFICATIONS

Military

MIL-T-81734(AS)	Test Set, Airborne Weapon Control System, AN/AWM-23(V)

DRAWINGS

Hughes Aircraft Company

493602-1XX	Adapter, Handling Antenna
493614-1XX	Adapter, Handling, TID
493619-1XX	Bar, Lifting, Antenna
493625-1XX	Adapter, Hoisting, Antenna

2.2 Availability of documents. Applicable (see 6.6)

2.3 Precedence of documents. When the requirements of the contract, this specification, or applicable subsidiary documents are in conflict, the following precedence shall apply:

a. Contract - The contract shall have precedence over any other document.

b. This specification - This specification shall have precedence over all applicable subsidiary documents (see 6.6).

c. Referenced documents - Any document referenced in this specification shall have precedence over all applicable documents referenced therein,

3. REQUIREMENTS

3.1 Materials, parts, and processes. Applicable (see 6.6).

3.1.1 Selection of materials, parts, and processes. Applicable.

3.1.1.1 Approval of nonstandard materials, parts, and processes, Applicable.

3.1.2 Flammable materials. Applicable.

3.1.3 Cables and connectors. Not applicable.

3.1.4 Wire. Not applicable.

3.1.5 Rotary switches. Not applicable

3.1.6 Finishes. Applicable.

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3.1.7 Marking and identification. Applicable.

3.1.8 Soldering. Not applicable.

3.1.9 Welding. Applicable.

3.1.10 Printed wiring. Not applicable.

3.1.11 Engineering changes to printed circuit boards. Not applicable.

3.1.12 Replaceability of electronic standard parts and materials. Not applicable.

3.2 Design and construction. Applicable.

3.2.1 Design features. Applicable.

3.2.1.1 Method of operation. Applicable.

3.2.1.2 Commercial test equipment. Not applicable.

3.2.1.2.1 Other commercial components. Not applicable.

3.2.1.3 Circuit grounds, Not applicable.

3.2.1.4 Interchangeability. Applicable.

3.2.1.5 Identification of product. Applicable.

3.2.1.6 Human factors. Applicable.

3.2.1.7 Test set generated acoustic noise. Not applicable.

3.2.1.8 Forced-liquid cooling. Not applicable.

3.2.1.9 Space and weight limitations. The maximum size and weight of the SHE shall be as listed in Table I.

3.2.1.10 Connectors. Not applicable.

3.2.1.11 Secondary electrical power. Not applicable.

3.2.1.12 Interface. The antenna handling adapter shall interface with the AN/AWG-9 radar antenna. The tactical information display handling adapter shall interface with the AN/AWG-9 tactical information display. The antenna lifting bar shall interface with the AN/AWG-9 radar antenna. The antenna hoisting adapter shall provide an interface between the hoist mechanism and the AN/AWG-9 radar antenna.

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3.2.2 Utilities. Not applicable.

3.2.3 Service conditions.

3.2.3.1 Electrical. Not applicable.

3.2.3.2 Environmental. Applicable.

TABLE I
SPECIAL HANDLING EQUIPMENT
FOR WEAPON REPLACEABLE ASSEMBLIES

Item	Nomenclature (see 6.4)	WRA Application (see 6.4)	Size (inches max)	Weight (lbs max)
1	Adapter, Handling Antenna MX-9046( )/ AWM-23(V)	Antenna AS-2764( )/ AWG-9	44 by 40 by 32	48
2	Adapter, Handling Tactical Information Display MX-9047( )/ AWM-23(V)	Display, Tactical IP-1110( )/AWG-9	31 by 17 by 21	20
3	Bar, Lifting, Antenna MX-9048( )/ AWM-23(V)	Antenna AS-2764( )/ AWG-9	48 by 39 by 25	19
4	Adapter, Hoisting, Antenna ADU-371( )/E	Antenna AS-2764( )/ AWG-9	13 by 10 by 8	15
5	Adapter, Handling, Antenna MX-9691( )/ AWM-23(V)	Antenna AS-2764( )/ AWG-9	44 by 40 by 33	48

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3.2.3.2.1 Organizational level. Applicable.

3.2.3.2.2 Intermediate level. Applicable.

3.2.4 Maintainability. Applicable.

3.2.4.1 Mean time to repair. Applicable.

3.2.4.2 Test points. Not applicable.

3.2.4.3 Shop replaceable assemblies. Not applicable.

3.2.4.4 Self test capabilities. Not applicable.

3.2.5 Reliability. Not applicable.

3.2.6 Electromagnetic Interference (EMI). Not applicable.

3.2.7 Utilities protection. Not applicable.

3.2.8 Calibration cycle. Not applicable.

3.3 Performance

3.3.1 General. The SHE shall facilitate the safe removal, installation, and transportation of the weapon replaceable assemblies (WRAs) listed in Table I at the organizational and intermediate maintenance levels.

3.3.2 Organization level functional requirements. Applicable.

3.3.3 Intermediate level functional requirements, Applicable.

3.3.4 Warm-up time. Not applicable.

3.4 Detailed functional requirements. Applicable.

3.5 Workmanship. Applicable.

3.6 SHE elements. Unless otherwise specified, the SHE shall be as specified in the following.

3.6.1 Handling Adapters. Handling adapters shall be welded assemblies that facilitate handling of, and provide physical protection for, the WRAs with which they are to be used. They shall be constructed to permit attachment to a suitable dolly by means of straps (not supplied with the adapters).

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3.6.1.1 Adapter, Handling, Antenna MX-9046( )/AWM-23(V) and MX-9691( )/AWM-23(V). The antenna handling adapter shall be so designed and constructed that two adapters may be attached to one another to provide a protective frame enclosing an antenna secured to one of the adapters. Each adapter shall provide a mounting surface with tapped holes located to match the mounting holes in the base of Antenna AS-2764( )/AWG-9. Hexagon headed cap screws (captive to the adapter) shall be provided to fasten the antenna to the adapter. Provision shall be made for mounting the adapter in Stand, Mounting, Antenna MT-4445 ( )/AWM-23(V). The adapter shall provide maintenance access to the sides, top, bottom, and rear of the radar antenna.

3.6.1.2 Adapter, Handling, Tactical Information Display MX-9047( )/AWM-23(V). The TID handling adapter shall provide protective mounting for the TID. Bolts and pins used for attachment shall be captive to the adapter.

3.6.1.3 Bar, Lifting, Antenna MX-9048( )/AWM-23(V). The lifting bar shall engage the antenna hoisting lugs to facilitate removal and installation of the radar antenna. The lifting bar shall provide for application of a lifting force acting along a vertical line through the center of gravity of the radar antenna.

3.6.1.4 Adapter, Hoisting, Antenna ADU-371( )/E. The antenna hoisting adapter shall provide captive attachment pins or screws for attachment of the adapter to the radar antenna. The adapter shall have an open jaw clevis with jaw spacing of one-half inch minimum and a captive (to the adapter) clevis pin with a maximum diameter of 3/8 inch for attachment to a hoist ring. The open jaw clevis shall be located to provide a lift point directly over the radar antenna center of gravity.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Applicable.

4.2 General. Applicable.

4.2.1 Classification. Applicable.

4.2.2 Individual items. Applicable.

4.2.3 In-process inspection. Applicable.

4.2.4 In-process data. Applicable.

4.2.5 Acceptance test plan. Applicable. Table II shall apply,

4.2.6 Test results. Applicable.

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TABLE II

ACCEPTANCE TEST PLAN FOR HANDLING EQUIPMENT

AN/AWM-23(V)

1.0 TEST OBJECTIVE. To demonstrate that the material, workmanship and mechanical interchangeability requirements of 3.1, 3.5, and 3.2.1.4, respectively, have been met.

2.0 TEST CONDITIONS (None applicable).

3.0 TESTS. Inspect each item listed below for conformance to the applicable assembly or outline and mounting drawing.

EQUIPMENT ITEMS

Name	Part No.
1. Adapter, Handling, Antenna	493602-1XX
2. Adapter, Handling, TID	493614-1XX
3. Adapter, Hoisting, Antenna	493625-1XX
4. Bar, Lifting, Antenna	493619-1XX

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4.2.7 Acceptance. Applicable.

4.2.8 Utilities, Not applicable.

4.2.9 Configuration. Applicable.

4.3 Tests.

4.3.1 Electrical and electronic items - operational run-in test. Not applicable.

4.3.2 Mechanical items. Applicable.

4.4 Environmental test conditions. Applicable.

4.5 Test location. Applicable.

4.6 Acceptance schedule. Applicable.

4.7 Test procedures. Applicable.

4.8 Rejection and retest. Applicable.

5. PREPARATION FOR DELIVERY

5.1 General. Applicable.

6. NOTES

6.1 Intended use. The SHE is intended to be used at organizational or intermediate levels to facilitate handling of certain WRAs used in Weapon Control System (WCS)AN/AWG-9.

6.2 Ordering data. Applicable.

6.3 Definitions. Applicable.

6.4 Nomenclature cross reference. The following is a cross reference between the official Navy nomenclature and the major contractor part numbers. The contractor part number suffix is controlled by the configuration list in the contract.

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	Official Nomenclature	Contractor Part Number
a.	Antenna AS-2764( )/AWG-9	481031-1XX
b.	Display, Tactical IP-1110( )/AWG-9	481580-1XX
c.	Stand, Mounting, Antenna MT-4445( )/AWM-23(V)	493354-1XX
d.	Adapter, Handling, Antenna MX-9046( )/AWM-23(V)	493602-1XX
e.	Adapter, Handling, Tactical Information Display MX-9047 ( )/AWM-23(V)	493614-1XX
f.	Bar, Lifting, Antenna MX-9048 ( )/AWM-23(V)	493619-1XX
g.	Adapter, Hoisting, Antenna ADU-371( )/E	493625-1XX
h.	Adapter, Handling, Antenna MX-9691( )/AWM-23(V)	493602-1XX
i.	Test Set, Antenna Boresight TS-3568( )/AWM-23(V)	493720-1XX

6.5 Liquid cooler. Not applicable.

6.6 Use of this specification. This specification is a detail specification which uses MIL-T-81734(AS) as a general specification. Where the corresponding paragraphs of MIL-S-81780(AS) are the same as MIL-T-81734(AS) the word “applicable” is used in this specification. When the term “not applicable” is used in this specification, it means that the corresponding paragraph requirements of MIL-T-81734(AS) are not required to be met (see 6.6 of MIL-T-81734(AS)).

6.7 Abbreviations and acronyms. Applicable. In addition the following shall apply:

SHE - Special Handling Equipment
TID - Tactical Information Display

6.8 Convenience outlet power. Not applicable.

6.9 Changes from previous issue are not marked.

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MIL-S-81780C(AS) Special Handling Equipment

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
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